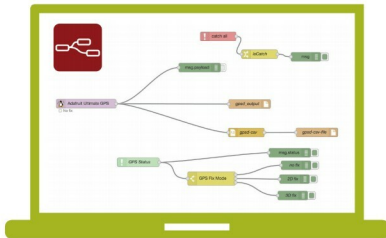


A Few Facts About Node-RED

Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways. It provides a browser-based editor.



Since early 2014, the mobile has overtaken the personal computer (desktop/laptop) as the leading device used to navigate the Net. Along with the mobile, a number of other portable devices that connect to the Internet have also started proliferating at a very quick rate. Nowadays, most of us carry or possess at least one Internet based device and a mobile. So, the Internet of Things (IoT) now doesn't only mean different 'things', but has evolved into 'intelligent things' which have on-board computation and network connections. Most importantly, they have the capability to sense the environment around us and, accordingly, act intelligently. These devices are now referred to as connected devices, smart objects or the Web of Things.

This adaptation of technology has led to many developments in the visual programming and visual coding arenas. As a result, Node-RED has evolved into a visual programming/coding tool based on the already popular Node.js (a server side JavaScript platform), mainly targeting the Internet of Things space.

What is Node-RED?

Node-RED is a programming tool for wiring together hardware devices, APIs and online services. Primarily, it is a visual tool designed for the Internet of Things, but it can also be used for other applications to very quickly assemble flows of various services.

Node-RED is open source and was originally created by the IBM Emerging Technology organisation. It is included in IBM's Bluemix (a Platform-as-a-Service or PaaS) IoT starter application package. Node-RED can also be deployed

separately using the Node.js application. At present, Node-RED is a JS Foundation project, about which more details can be found at <https://nodered.org/>.

Node-RED enables users to stitch together Web services and hardware by replacing common low-level coding tasks (like a simple service talking to a serial port), and this can be done with a visual drag-drop interface. Various components in Node-RED are connected together to create a flow. Most of the code needed is created automatically.

Installation of Node-RED

To start using Node-RED, we have to first install it. This task is platform-specific and we have to choose the right installable for the respective platform. To make Node-RED work, we need to install Node.js. At present the recommended versions of Node.js are LTS 6.x or 6.x.

- 1) To install and configure Node-RED, first install the required Node.js, as follows:

```
$ sudo apt-get install node.js
```

- 2) Once you are done with the Node.js set-up, install the Node Package Manager (npm) as follows:

```
$ sudo apt-get install npm
```

- 3) After npm configuration is done, create a symbolic link with 'node' for 'node.js' as shown below:

```
$ ln -s /usr/bin/nodejs /usr/bin/node
```